Data Structures for Transaction

- A transaction is a data structure that encodes a transfer of value from a source of funds, called an input, to a destination, called an output.
- The Version field specifies the rules that this transaction follows.
- The Input Counter field indicates the number of inputs included in the transaction, while the Input fields contain one or more transaction inputs.
- The Output Counter field denotes the number of outputs included, and the Output fields contain one or more transaction outputs.
- The Lock Time field represents a Unix timestamp or block number, which serves as a condition that must be met before the transaction can be added to the blockchain.

Field	Description
Version	Transaction version number
Input Counter	Number of inputs in transaction
Input 1	Input details for first input
Input 2	Input details for second input
...	...
Output Counter	Number of outputs in transaction
Output 1	Output details for first output
Output 2	Output details for second output
...	...
Lock Time	Transaction lock time

Types of Transactions

Different types of transactions in the Bitcoin system have different functions and characteristics.

Standard Transaction:

A standard transaction entails an uncomplicated exchange of value between two parties. It consists of inputs that refer to prior transaction outputs as the source of funds and outputs that assign the transferred value to new recipients.

Multisignature Transaction:

A multisignature transaction necessitates the approval of multiple parties prior to the expenditure of funds. Multiple public keys, often pertaining to distinct individuals or entities, are utilized as inputs for the transaction. Typically, the outputs of the transaction are then locked with a script that specifies the required number of signatures to expend the funds.

Types of Transactions

Segregated Witness (SegWit) Transaction: Segregated Witness is an upgrade to the Bitcoin protocol that isolates the transaction signature data (witness) from the actual transaction data. SegWit transactions have a unique structure and use a unique transaction format, allowing for an increase in transaction capacity and enhanced scalability.

Coinbase Transaction: A Coinbase transaction is a unique type of transaction in Bitcoin that creates new bitcoins and awards them to the miner who successfully adds a new block to the blockchain. It is the first transaction in every block and serves as the mechanism for issuing new bitcoins into circulation, as well as rewarding miners for their work in securing the network.

Lock Time–Enabled Transaction: Lock time enables the sender of a transaction to specify a future time or block height at which the transaction will become valid. Lock time–enabled transactions are useful for a variety of use cases, including the creation of time-locked transactions, the implementation of conditional spending, and the creation of payment channels similar to the Lightning Network.

Types of Transactions

❑ Segregated Witness (SegWit) Transaction

Segregated Witness (SegWit) is an **upgrade to the Bitcoin protocol** introduced in **2017** (BIP 141).

It solves two major problems in Bitcoin:

Transaction malleability (a bug that allowed modification of transaction IDs)

Limited block capacity (only about 1 MB of data per block)

A **Bitcoin transaction** has two main parts:

Transaction Data – who sends how many bitcoins to whom

Signature (Witness) Data – cryptographic proof that the sender owns those bitcoins

Before SegWit, both of these were **stored together inside the block**.

SegWit **separates (segregates)** the **witness (signature)** from the main transaction data.

This means the **signature data is moved to a separate structure**, not counted fully toward the 1 MB block limit.

Types of Transactions

❖ **Example:** Before vs After SegWit

❑ **Before SegWit (Legacy Transaction)**

Transaction ID

Inputs (with signatures) | <-- Signatures included here

Outputs (who receives bitcoins)

- All data including signatures is inside the same structure.
- So, if a transaction had large signatures, it made the block big quickly.

❑ **After SegWit**

Transaction ID

Inputs (without signatures) | <-- Smaller now

Outputs

↓

Separate “Witness Data”

(Signatures stored here)

Now the **witness (signature)** part is stored separately, reducing the “main” transaction size and allowing **more transactions per block**.

Types of Transactions

❑ Illustrative Example

Let's say:

Alice sends **1 BTC** to Bob.

The transaction has **250 bytes** of data.

Out of that, **100 bytes** are signatures.

➤ **Before SegWit:**

The entire 250 bytes count toward the block size limit (1 MB).

➤ **After SegWit:**

Only **150 bytes (non-witness data)** count fully toward the 1 MB limit.

The **100 bytes of witness data** count at a **reduced weight** (1/4 of normal).

Effective total = 150 + (100 / 4) = **175 bytes**.

This means more transactions fit into one block.

Types of Transactions

❑ Benefits of SegWit

Increased capacity: Because witness data is discounted, more transactions fit in a block.

Lower fees: Smaller effective size = less fee per transaction.

Fixes malleability: Transaction IDs no longer change if the signature changes.

Enables Layer 2 solutions: Technologies like the **Lightning Network** depend on SegWit.

❑ **Legacy address VS. SegWit address**

A **legacy address** starts with 1

A **SegWit address** starts with 3 (P2SH) or bc1 (bech32).

Example:

Legacy Address: 1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa

SegWit Address: bc1qar0srrr7xfkvy5l643lydnw9re59gtzzwf5mdq

Types of Transactions

- **A real life Analogy:**
- Think of a **Bitcoin block** as a **suitcase.**
- Before SegWit, you packed **everything (money + documents)** inside it, and it filled up quickly.
- After SegWit, you moved the **documents (signatures)** to a **side pouch**, freeing up more room for **actual items (transactions).**
- More fits in one suitcase = more transactions per block!

Types of Transactions

What is Locktime?

Locktime (nLockTime) is a field in every Bitcoin transaction that specifies **when the transaction becomes valid** — meaning **it can't be included in a block until a specific time or block height** has passed.

So, **Locktime = "Do not spend this transaction before ___."**

How It Works Internally

By block height

- If nLockTime < 500,000,000, it represents a **block height.**
- Example: nLockTime = 800000 → valid only after block #800,000 is mined.

By Unix timestamp

- If nLockTime >= 500,000,000, it represents a **Unix time** (seconds since 1 Jan 1970).
- Example: nLockTime = 1759872000 → valid only after **October 8, 2025, 00:00:00 UTC.**

Types of Transactions

Example: Time-Locked Savings Account

Alice wants to send **1 BTC to her son**, but she wants him to receive it **on his 18th birthday**, not before.

She creates a transaction like this:

Field	Value
Input	Alice's UTXO
Output	1 BTC → Son's address
nLockTime	**October 8, 2030 (Unix time = 1917638400)**

Meaning:

Even if someone broadcasts this transaction today, it will **not be mined** into a block until after **October 8, 2030**.

Types of Transactions

What is Block Height?

Block height refers to the **position or number of a block in the blockchain**. It represents **how many blocks have been added to the blockchain since the genesis block (the first block).**

For example:

- The **genesis block** (first block) has a block height of **0**.
- The next block has a height of **1**, and so on.

For example, if a transaction has a **lock time of block height 800,000**, it means:

- The transaction **cannot be included in the blockchain until block 800,000 is mined**.
- If the current block height is **799,500**, the transaction remains **invalid** until 500 more blocks are added.

Types of Transactions

Why is Lock Time is useful?

Lock time enables **time-locked transactions**, allowing:

- **Delayed payments** (e.g., escrow releases at a future date).
- **Conditional spending** (funds become available only after a certain block height).

Transaction Input and Output

- Unspent transaction outputs (UTXOs) are the foundational building block of a bitcoin transaction.
- UTXOs are immutable units of bitcoin currency that are locked to a specific proprietor, recorded on the blockchain, and recognized by the entire network as currency units.
- The bitcoin network keeps track of the millions of available (unspent) UTXOs. Every time a user receives bitcoin, the quantity is recorded as a UTXO in the blockchain.
- A user's bitcoins could be dispersed as UTXO across hundreds of transactions and hundreds of blocks.
- In practice, there is no such thing as a Bitcoin address or account balance; only scattered UTXOs are locked to specific proprietors.
- The wallet application generates the notion of a user's Bitcoin balance. The wallet computes the user's balance by searching the blockchain and summing all of the user's UTXOs.

Transaction Input and Output

- Once created, UTXOs must be consumed in their entirely and generate change. For example, a transaction with a 20-Bitcoin UTXO would consume the entire 20 Bitcoin UTXO and produce two outputs: paying 1 Bitcoin to the recipient and returning 19 Bitcoins in change back to the wallet.
- Most bitcoin transactions generate change. A user's wallet application automatically selects various units from their available UTXOs to compose an amount greater than or equal to the desired transaction amount.
- The wallet application can use strategies like combining smaller units, finding exact change, or using a single unit larger than the transaction value and making change.
- Transaction inputs are the UTXOs consumed by a transaction, while outputs are the UTXOs created by a transaction.

Mining Bitcoin

- Mining is the process of adding new transactions to a blockchain and securing the network through the resolution of complex mathematical riddles.
- Network participants, known as miners, compete to solve a cryptographic algorithm known as proof of work (PoW).
- Miners collect valid transactions into blocks and then apply a mathematical function (hashing) to the block's data, which includes a reference to the previous block.
- The newly mined block is broadcast to the network by the first miner to discover a valid hash satisfying the criteria. Other miners then verify the block's transactions for validity.
- If consensus is achieved that the block is valid, it is added to the blockchain and the miner is compensated with newly minted Bitcoins and transaction fees from the included transactions.

Mining Bitcoin

Bitcoin mining serves the two following primary purposes:

- **Transaction validation:** Miners ensure the integrity and accuracy of the Bitcoin ledger by validating and including valid transactions in new blocks.
- **Blockchain security:** As a security mechanism, the computational effort required for mining (PoW) serves as a security measure.

Nakamoto consensus is the consensus mechanism utilized by Bitcoin. Nakamoto consensus is attained through the consensus of the majority of miners. Miners determine collectively which chain is deemed "longest" and, therefore, legitimate. This is determined by the chain with the most accumulated PoW, indicating the greatest investment of computational effort in its creation.

Cost of Mining Bitcoin

Bitcoin mining incurs substantial costs due to the required resources and associated expenses. What follows is a summary of the costs associated with Bitcoin mining.

- **Hardware Costs:** Bitcoin mining necessitates mining rigs or Application-Specific Integrated Circuits (ASICs). These machines are designed specifically to execute the computational tasks required for mining.
- **Electricity Costs:** The mining industry consumes a considerable quantity of electricity. Mining necessitates a significant amount of computational capacity, resulting in substantial electricity costs. Miners must consider the local electricity rates and calculate the energy consumption of their mining apparatus in order to estimate their ongoing costs.
- **Cooling and Maintenance Costs:** The intense computational activities of mining equipment generate heat. Effective cooling systems are required to prevent combustion and prolong the life of mining machinery. These cooling solutions, such as fans and specialized cooling systems, require additional expenditures.

Cost of Mining Bitcoin

- **Operating Costs:** In addition to hardware and electricity costs, mining incurs additional operating expenses. Expenses associated with Internet connectivity, facility rental (if mining operations are conducted in a dedicated space), mining software licenses, and other miscellaneous expenses may be included.
- **Competition and Return on Investment:** The level of competition influences the expense of Bitcoin mining. As more miners join the network, the mining difficulty rises, necessitating more computational power and, as a result, increased electricity consumption. Individual miners' return on investment (ROI) is diminished as a result of escalating competition.
- **Market Volatility:** Bitcoin's price volatility is an additional factor that impacts mining costs. The fluctuating price of Bitcoin has a direct impact on the profitability of Bitcoin mining. A higher Bitcoin price can make mining more profitable, whereas a significant decline in price can make mining less lucrative or even unprofitable for some miners.

51% Attack in Bitcoin Mining

- A 51% attack occurs when a single entity or group of cooperating entities controls more than fifty percent of the network's total mining capacity.
- This majority control enables the attacker to potentially control the blockchain and dictate the norms of consensus.
- The adjacent table delineates the definition of a 51% attack within the realm of cryptocurrency. The provided explanation elucidates that a 51% attack transpires when an entity acquires dominion over more than 50% of the mining power within a network.

Aspect	Description
Definition	Control threshold: Over 50% of network's mining power
Consequences	Double-spending Network manipulation Erosion of trust
Prevention	Network security Decentralization Vigilance
Past incidents	Mt. Gox (2014) - Bitcoin Ethereum Classic (2020) - Ethereum Classic Verge (2018) - Verge Feathercoin (2013) - Feathercoin

How to Mitigate 51% Attack

Although a 51% attack is theoretically possible, several factors mitigate the risks:

Incentives for honesty: There are economic incentives for honest behavior and adherence to consensus norms among miners. Any attempt to manipulate the network could erode trust, decrease Bitcoin's value, and harm the attacker's own investments.

Decentralization: The decentralized nature of cryptocurrency mining guarantees that no single entity will be able to control the entire network. This decentralization makes acquiring majority control of mining power difficult and expensive.

Community vigilance: The Bitcoin community monitors the network for suspicious activity. Rapid detection and disclosure of potential attacks detect criminals.

Network upgrades: Periodic Bitcoin protocol updates and enhancements resolve known vulnerabilities and strengthen the network's resistance to attack.